

CSA0310 – DATA STRUCTURES

ASSESSMENT TOOL-CO3 -AT3

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Challenge 13 – Student/roll no.: _____



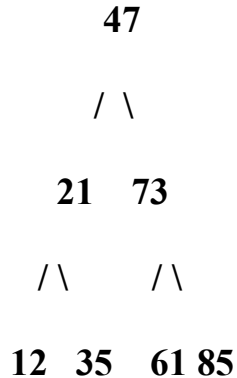
A. Binary Tree (4 marks): Write the preorder, inorder, postorder and level-order traversals of the given tree.

B. BST (3 marks): Insert the keys in this order: 47, 21, 73, 11, 34, 62, 85. Draw the final BST and write its postorder traversal.

C. AVL Tree (3 marks): Insert 60, 30, 40 into an empty AVL tree. Name the rotation, draw the balanced tree and write its preorder traversal.

Part A: Binary Tree Traversals

Given Binary Tree:



1. Preorder Traversal (Root → Left → Right)

Answer: 47, 21, 12, 35, 73, 61, 85

2. Inorder Traversal (Left → Root → Right)

Answer: 12, 21, 35, 47, 61, 73, 85

3. Postorder Traversal (Left → Right → Root)

Answer: 12, 35, 21, 61, 85, 73, 47

4. Level-order Traversal (Level by Level)

Answer: 47, 21, 73, 12, 35, 61, 85

PART B. BST

Insert keys in this order:

47, 21, 73, 11, 34, 62, 85

Final BST:

47

**/ **

21 73

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11 34 62 85

Postorder Traversal:

11, 34, 21, 62, 85, 73, 47

PART C. AVL Tree

Insert: 60, 30, 40

Step 1: Insert 60

60

Step 2: Insert 30

60

/

30

Step 3: Insert 40

60

/

30

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40

This creates a Left-Right (LR) Rotation case.

Rotation Name: Left-Right (LR) Rotation

Balanced AVL Tree:

40

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30 60

Preorder Traversal:

40, 30, 60